

Force Reconstruction Algorithm (Version 1.4)

Input:

- $n_{sensors}$ (number of sensing channels)
- $N_W = 1000$ (integration window length)
- $Thr_{samples} = 1500$ (noise learning samples)
- $n_{\sigma} = 10.0$ (threshold scale factor)
- $press_{confirm} = 5$ (trigger confirmation samples)
- $fifo_{len} = 20$ (release slope FIFO length)
- $avg_{multiplier} = 0.5$ (release slope multiplier)
- $\alpha = 1$ (EMA smoothing factor)
- $n_{offset} = 50$ (adaptive offset window)
- $samples_{after_release} = 500$ (guard samples)
- $signal2noise = 10.0$ (minimum SNR)
- $min_{press_sensors} = 4$ (global press quorum)
- $release_{ratio} = 0.5$ (global release ratio)

Initialization

```
noise_counter  $\leftarrow$  0
noise_buffer[ $Thr_{samples}, n_{sensors}$ ]  $\leftarrow$  0
adaptive_offset[ $n_{sensors}$ ]  $\leftarrow$  0
offset_buffer[ $n_{offset}, n_{sensors}$ ]  $\leftarrow$  0
pre_buffer[ $press_{confirm} + 5, n_{sensors}$ ]  $\leftarrow$  0
press_buffer[ $N_W, n_{sensors}$ ]  $\leftarrow$  0
fifo_buffer[ $fifo_{len}, n_{sensors}$ ]  $\leftarrow$  0
 $x_{prev}[n_{sensors}] \leftarrow 0$ 
for  $ns = 1$  to  $n_{sensors}$  do
    integral[ $ns$ ]  $\leftarrow$  0
    counter[ $ns$ ]  $\leftarrow$  0
    confirm[ $ns$ ]  $\leftarrow$  0
    fifo_idx[ $ns$ ]  $\leftarrow$  0
    triggered[ $ns$ ]  $\leftarrow$  False
    validated[ $ns$ ]  $\leftarrow$  False
    armed[ $ns$ ]  $\leftarrow$  True
    press_sign[ $ns$ ]  $\leftarrow$  +1
    max_post[ $ns$ ]  $\leftarrow$  0
    second_cross[ $ns$ ]  $\leftarrow$  False
    idx_second_cross[ $ns$ ]  $\leftarrow$  0
    press_integral[ $ns$ ]  $\leftarrow$  0
    average_touch[ $ns$ ]  $\leftarrow$  0
    guard_counter[ $ns$ ]  $\leftarrow$  0
    previous_event_polarity[ $ns$ ]  $\leftarrow$  0
global_press  $\leftarrow$  False
press_mask[ $1..n_{sensors}$ ]  $\leftarrow$  False
release_mask[ $1..n_{sensors}$ ]  $\leftarrow$  False
1. Noise Learning Phase
for  $k = 1$  to  $Thr_{samples}$  do
```

```

└ Store  $x_{raw}[k, 1..n_{sensors}]$  into noise_buffer
 $\sigma[ns] \leftarrow \text{std}(\text{noise\_buffer}[:, ns])$ 
 $\text{thr}_{press}[ns] \leftarrow \text{press}_{\sigma} \cdot \sigma[ns]$ 
 $\text{max}_{noise}[ns] \leftarrow 99^{th}\%(|\text{noise\_buffer}[:, ns] - \text{median}|)$ 
Initialize offset_buffer and adaptive_offset

```

2. Online Processing Loop

```

for each new sample  $x_{raw}[1..n_{sensors}]$  do
  for  $ns = 1$  to  $n_{sensors}$  do
     $x_{smooth} \leftarrow \alpha x_{raw}[ns] + (1 - \alpha)x_{prev}[ns]$ 
     $x_{prev}[ns] \leftarrow x_{smooth}$ 
     $x \leftarrow x_{smooth} - \text{adaptive\_offset}[ns]$ 
    if  $\text{guard\_counter}[ns] > 0$  then
       $\text{guard\_counter}[ns] \leftarrow \text{guard\_counter}[ns] - 1$ 
      Output 0
    └ continue
    if  $|x| < 10 \cdot \text{thr}_{press}[ns] / \text{press}_{\sigma}$  then
      Update rolling mean of offset_buffer[:, ns]
      Update adaptive_offset[ns]
    Shift pre_buffer[:, ns], append  $x$ 

```

2.1 Trigger Detection

```

if not triggered[ns] then
  if armed[ns] and  $|x| > \text{thr}_{press}[ns]$  then
     $\text{confirm}[ns] \leftarrow \text{confirm}[ns] + 1$ 
    if  $\text{confirm}[ns] \geq \text{press}_{confirm}$  then
      triggered[ns]  $\leftarrow$  True
      armed[ns]  $\leftarrow$  False
       $\text{confirm}[ns] \leftarrow 0$ 
      second_cross[ns]  $\leftarrow$  False
       $\text{idx}_{second\_cross}[ns] \leftarrow 0$ 
       $\text{press\_sign}[ns] \leftarrow \text{sign}(\sum \text{pre\_buffer}[:, ns])$ 
      if  $\text{press\_sign}[ns] = 0$  then
        └  $\text{press\_sign}[ns] \leftarrow \text{sign}(x)$ 
       $\text{integral}[ns] \leftarrow \sum \text{press\_sign}[ns] \cdot \text{pre\_buffer}[:, ns]$ 
       $\text{press\_buffer}[0 : \text{pre}, ns] \leftarrow \text{cumsum}(\text{press\_sign}[ns] \cdot \text{pre\_buffer}[:, ns])$ 
      counter[ns]  $\leftarrow$  pre
    else
      └  $\text{confirm}[ns] \leftarrow 0$ 
      Output 0
    └ continue

```

2.2 Integration

```

 $\text{integral}[ns] \leftarrow \text{integral}[ns] + \text{press\_sign}[ns] \cdot x$ 
if  $\text{integral}[ns] < 0$  then
  Reset local state, enable guard
  Output 0
└ continue
 $\text{max}_{post}[ns] \leftarrow \max(\text{max}_{post}[ns], |x|)$ 
if  $\text{counter}[ns] < N_W$  then
   $\text{press\_buffer}[\text{counter}[ns], ns] \leftarrow \text{integral}[ns]$ 
   $\text{counter}[ns] \leftarrow \text{counter}[ns] + 1$ 
  Output 0
└ continue

```

Output $\text{integral}[ns]$

2.3 Second Threshold Crossing

```

if  $|x| < \text{thr}_{press}[ns]$  and not second_cross[ns] then
   $\text{idx}_{second\_cross}[ns] \leftarrow \text{counter}[ns]$ 

```

```

└─  $press\_integral[ns] \leftarrow integral[ns]$ 
└─  $second\_cross[ns] \leftarrow True$ 
2.4 Window Validation
if not validated[ns] then
     $idx \leftarrow \arg \max press\_buffer[0 : idx_{second\_cross}, ns]$ 
     $avg \leftarrow \frac{-press\_buffer[0, ns] + press\_buffer[idx, ns]}{idx + 1}$ 
    if  $max_{post}[ns] < signal2noise \cdot max_{noise}[ns]$  then
        Reset local state (no guard)
    └─ continue
     $average_{touch}[ns] \leftarrow avg$ 
     $validated[ns] \leftarrow True$ 
└─  $press\_mask[ns] \leftarrow True$ 
2.5 Global Press Consensus
if not global_press then
    if  $\sum press\_mask \geq min_{press\_sensors}$  then
        └─  $global\_press \leftarrow True$ 
2.6 Release Detection (FIFO Slope)
if validated[ns] and global_press then
    Update  $fifo\_buffer[:, ns]$ 
    if FIFO full then
         $slope \leftarrow \frac{fifo[-1] - fifo[0]}{fifo_{len}}$ 
        if  $slope < -slope_{mult}|average_{touch}[ns]|$  or  $integral[ns] > 1.5 press\_integral[ns]$  or  $integral[ns] < 0.5 press\_integral[ns]$  then
            └─  $release\_mask[ns] \leftarrow True$ 
2.7 Global Release
if  $\sum release\_mask / \sum press\_mask \geq release_{ratio}$  then
    Reset all sensors with guard enabled
     $press\_mask[:] \leftarrow False$ 
     $release\_mask[:] \leftarrow False$ 
     $global\_press \leftarrow False$ 

```